

● EASY

● ADVANCED MATH

Nonlinear equations in one variable and systems of equations in two variables

04

10 questions · ~15 min

$$x = 8$$

$$y = x^2 + 8$$

The graphs of the equations in the given system of equations intersect at the point x, y in the xy -plane. What is the value of y ?

- A. 8
- B. 24
- C. 64
- D. 72

$$\frac{x^2}{25} = 36$$

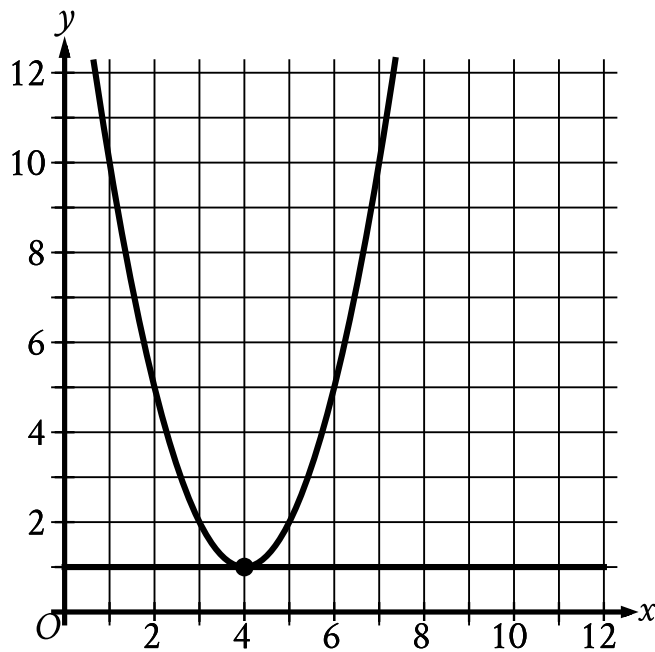
What is a solution to the given equation?

- A. 6
- B. 30
- C. 450
- D. 900

$$8x^2 - 40 = 32$$

What is the positive solution to the given equation?

- A. 3
- B. 4
- C. 9
- D. 72



- For the parabola in the system:
 - The parabola opens upward.
 - The vertex is at point (4 comma 1).
 - The parabola passes through the following points:
 - (3 comma 2)
 - (4 comma 1)
 - (5 comma 2)
- For the line in the system:
 - The line is horizontal.
 - The line passes through the following points:
 - (0 comma 1)
 - (4 comma 1)

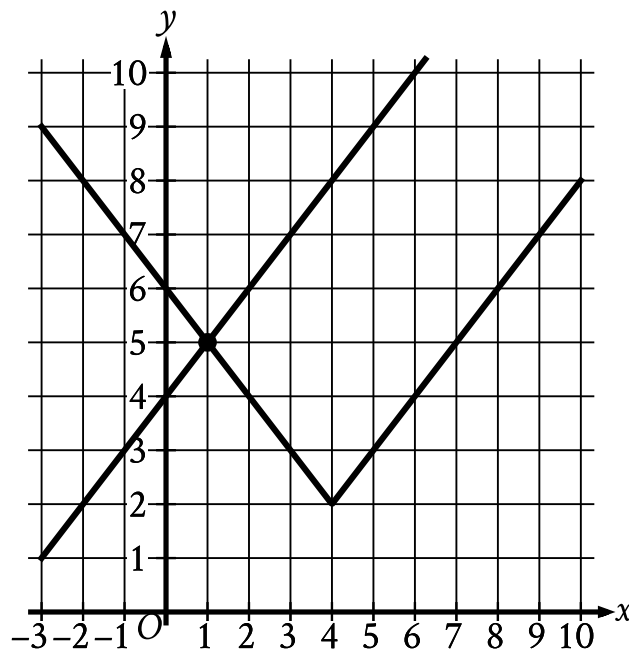
The graph of a system of a linear and a quadratic equation is shown. What is the solution x, y to this system?

A. 0,0

B. $-4, 1$

C. $4, -1$

D. $4, 1$



- For the absolute value function:
 - Moving from left to right:
 - The function slants sharply down to the point (4 comma 2).
 - The function then slants sharply up.
 - The function passes through the following points:
 - (1 comma 5)
 - (4 comma 2)
 - (6 comma 4)
- For the linear function:
 - The function slants sharply up from left to right.
 - The function passes through the following points:
 - (0 comma 4)
 - (1 comma 5)

The graph of a system of an absolute value function and a linear function is shown. What is the solution x, y to this system of two equations?

A. -1,5

B. 0,4

C. 1,5

D. 4,2

$$y - 57 = px$$

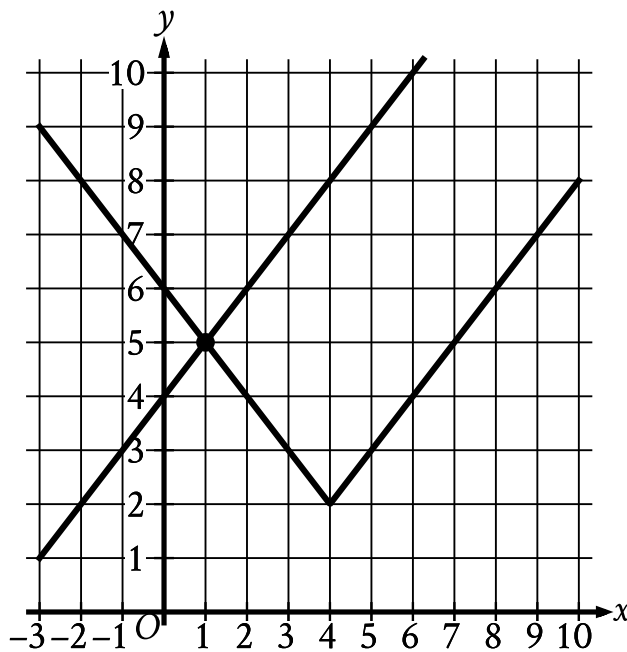
The given equation relates the positive numbers p , x , and y . Which equation correctly expresses y in terms of p and x ?

A. $y = 57x + p$

B. $y = px + 57$

C. $y = 57px$

D. $y = \frac{px}{57}$



- For the absolute value function:
 - Moving from left to right:
 - The function slants sharply down to the point (4 comma 2).
 - The function then slants sharply up.
 - The function passes through the following points:
 - (1 comma 5)
 - (4 comma 2)
 - (6 comma 4)
- For the linear function:
 - The function slants sharply up from left to right.
 - The function passes through the following points:
 - (0 comma 4)
 - (1 comma 5)

The graph of a system of an absolute value function and a linear function is shown. What is the solution x, y to this system of two equations?

A. -1,5

B. 0,4

C. 1,5

D. 4,2

$$x^2 = 64$$

Which of the following values of x satisfies the given equation?

- A. -8
- B. 4
- C. 32
- D. 128

Q10

GRID-IN

ADVANCED MATH

$$\sqrt{x+4} = 11$$

What value of x satisfies the equation above?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.